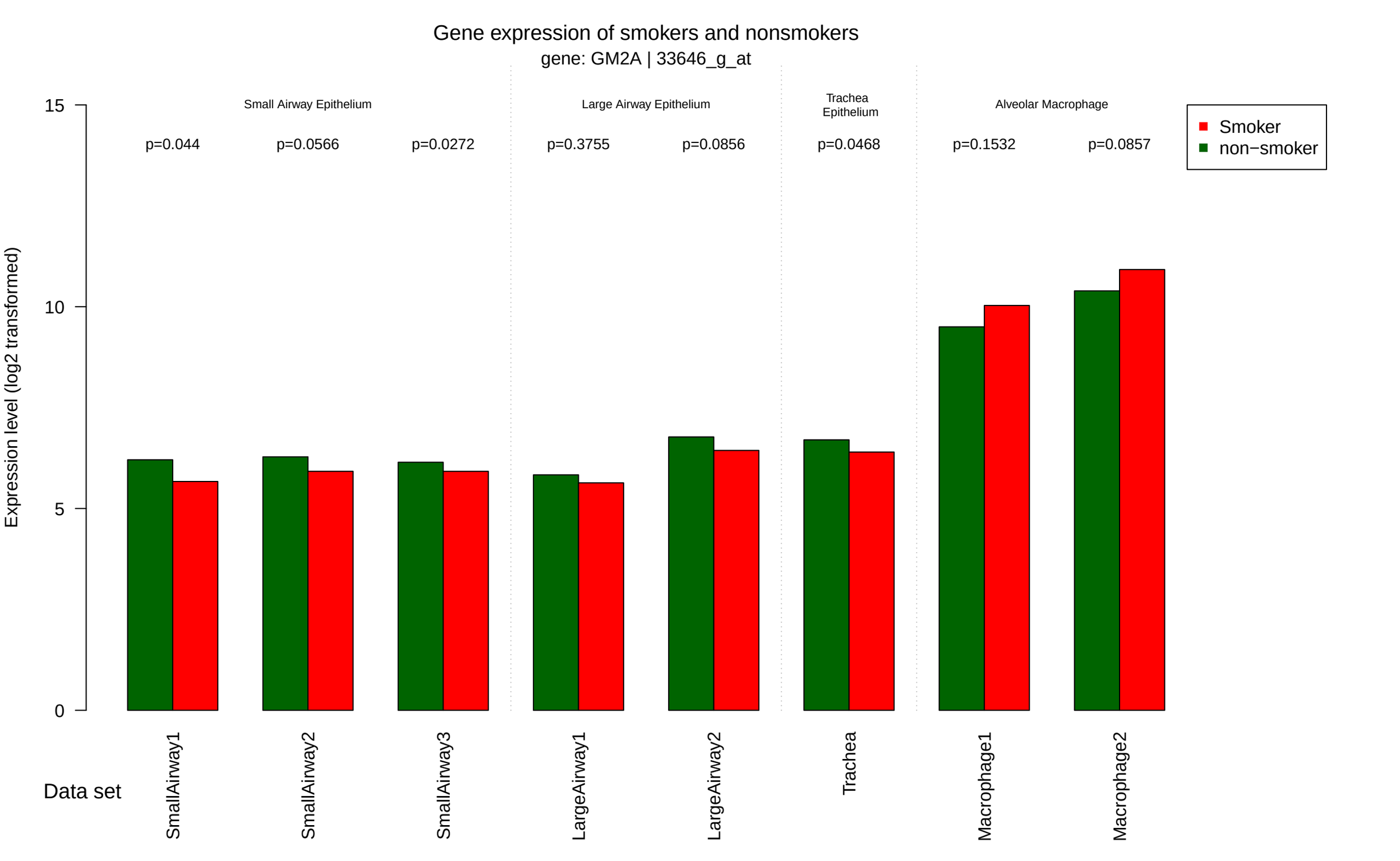
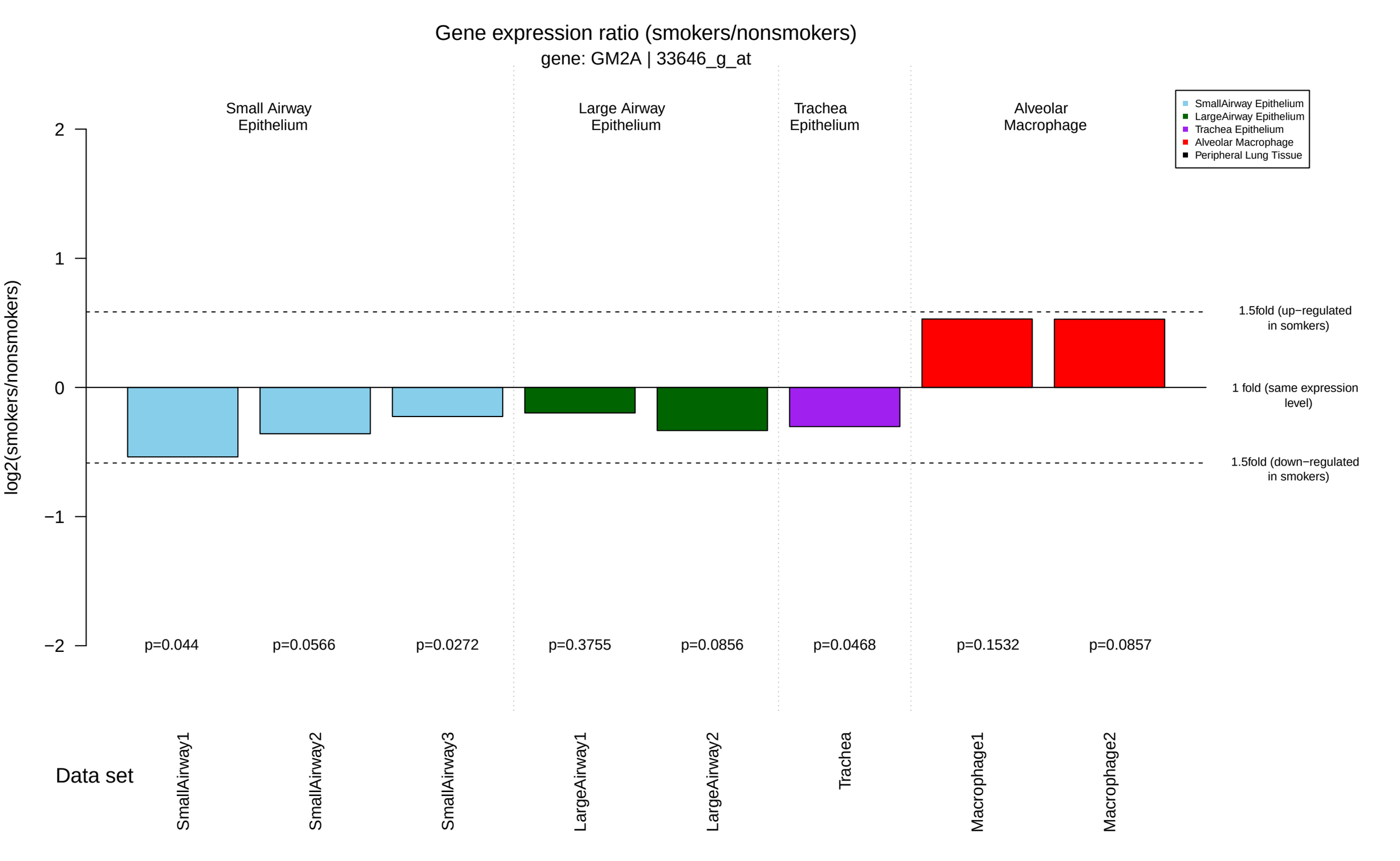
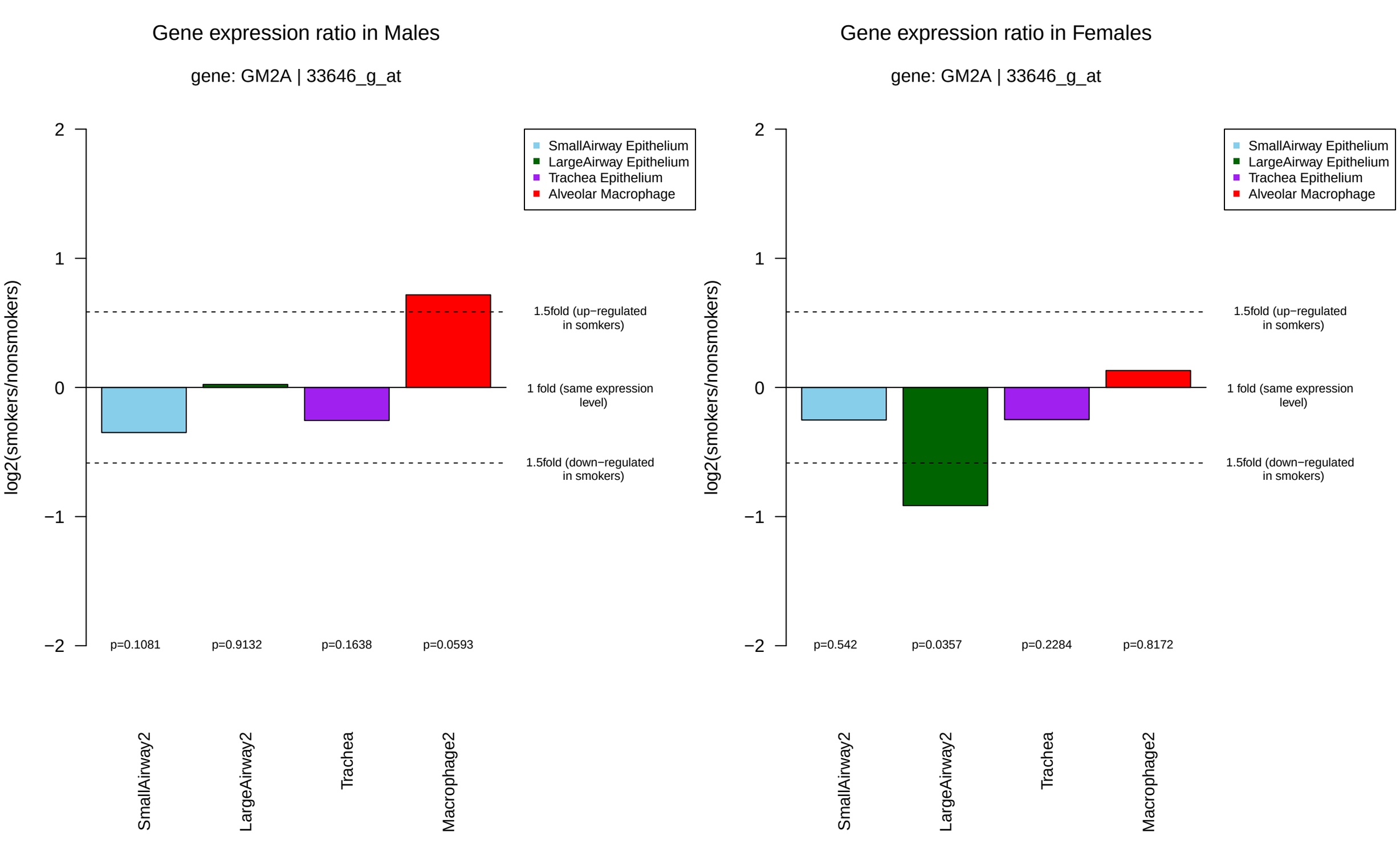
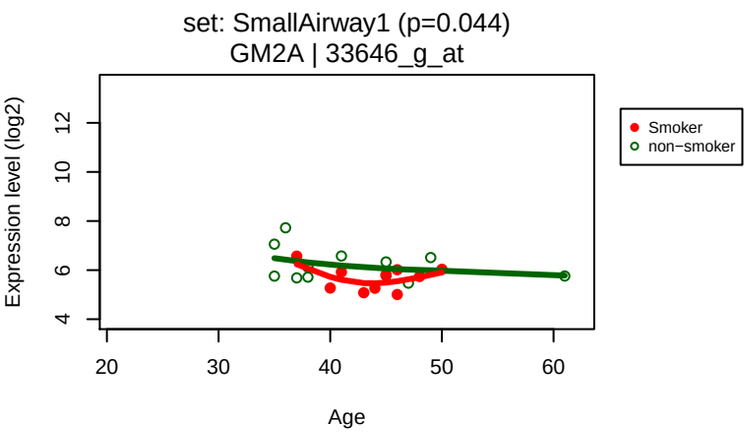
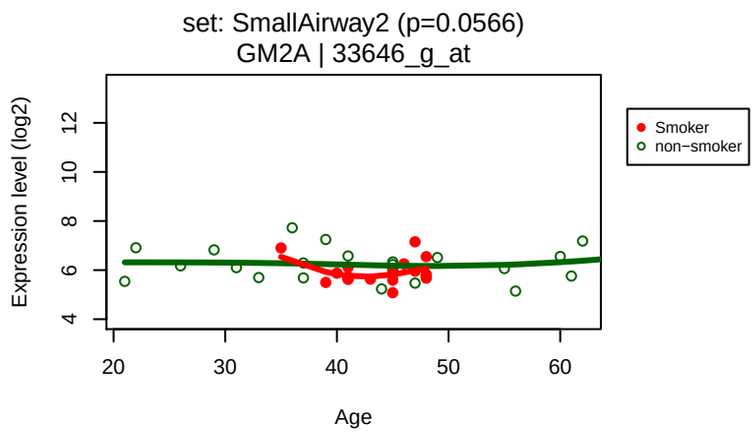
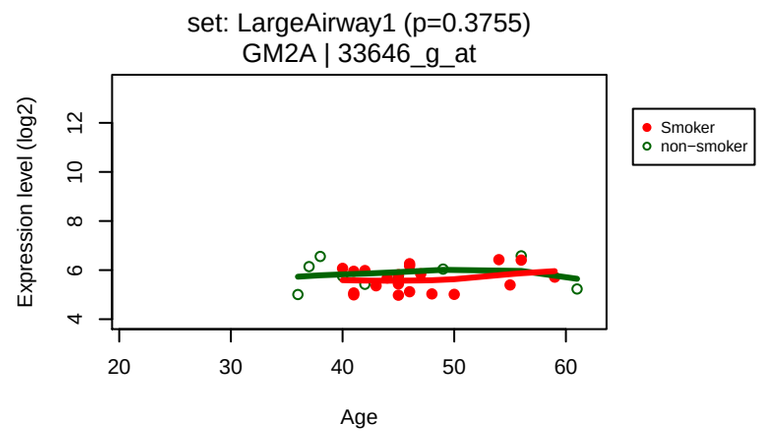

Gene expression of smokers and nonsmokers
gene: GM2A | 33646_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.044
p=0.0566
p=0.0272
p=0.3755
p=0.0856
p=0.0468
p=0.1532
p=0.0857
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

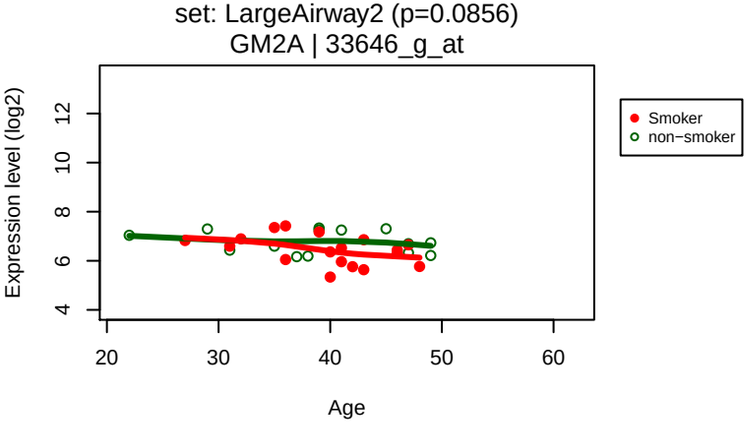

Gene expression ratio (smokers/nonsmokers)
gene: GM2A | 33646_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.044
p=0.0566
p=0.0272
p=0.3755
p=0.0856
p=0.0468
p=0.1532
p=0.0857
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

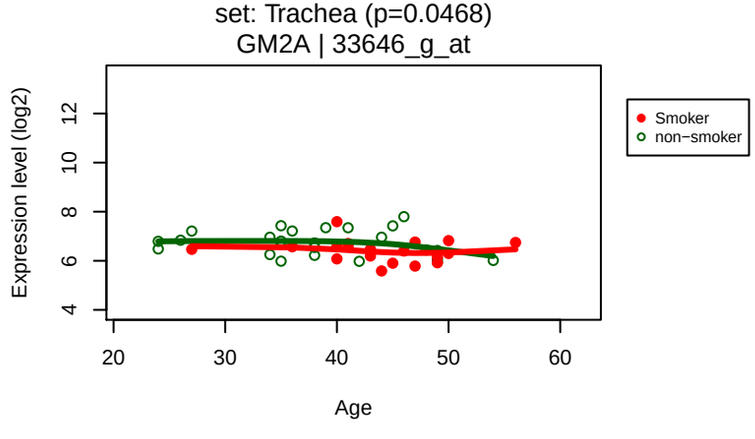

Gene expression ratio in Males
gene: GM2A | 33646_g_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1081
p=0.9132
p=0.1638
p=0.0593
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GM2A | 33646_g_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.542
p=0.0357
p=0.2284
p=0.8172
SmallAirway2
LargeAirway2
Trachea
Macrophage2

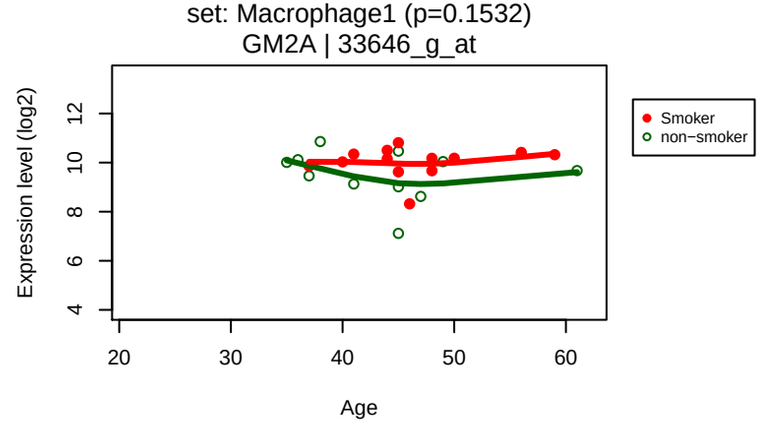

set: SmallAirway1 (p=0.044)
GM2A | 33646_g_at
Expression level (log2)
Age
Smoker
non−smoker

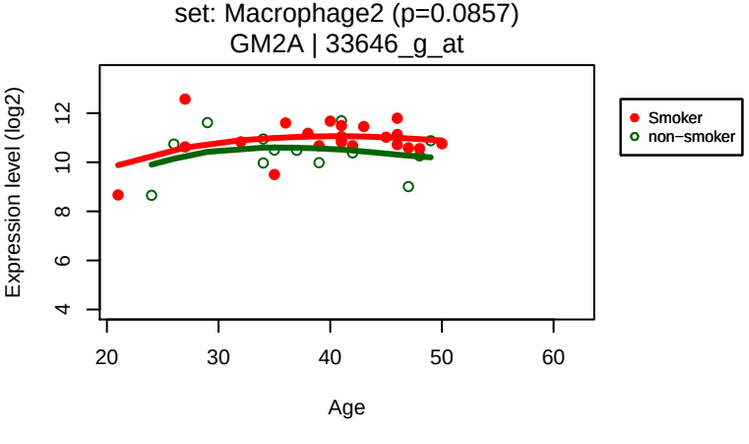

set: SmallAirway2 (p=0.0566)
GM2A | 33646_g_at
Expression level (log2)
Age
Smoker
non−smoker

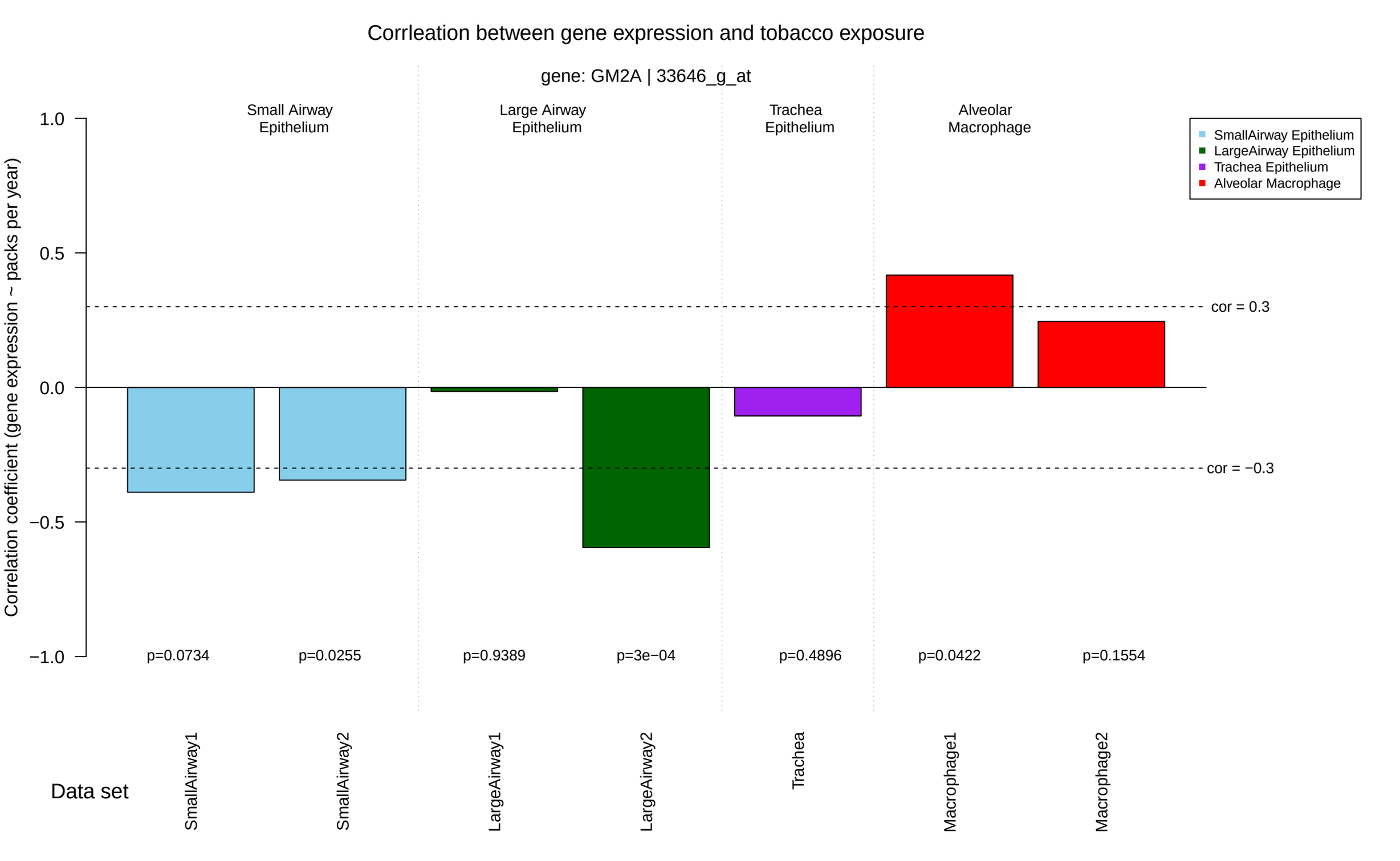

set: LargeAirway1 (p=0.3755)
GM2A | 33646_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0856)
GM2A | 33646_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0468)
GM2A | 33646_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1532)
GM2A | 33646_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0857)
GM2A | 33646_g_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GM2A | 33646_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0734
p=0.0255
p=0.9389
p=3e−04
p=0.4896
p=0.0422
p=0.1554
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2